

simulate_data

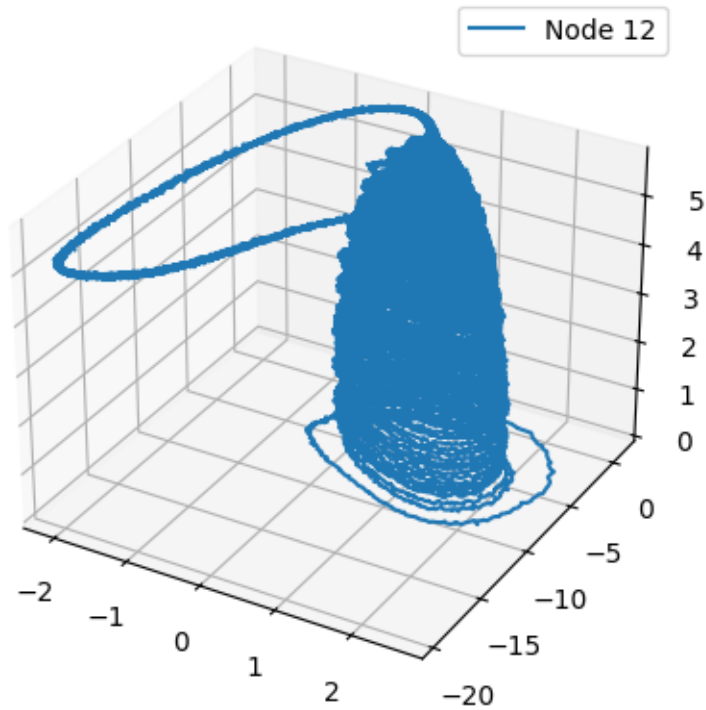
June 15, 2026

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
[2]: data = pd.read_csv('HR_timeseries_BA_deg_7_1_SNR_45.csv')
adj_matrix = pd.read_csv('BA_Nnodes100_Adj_deg_7_1.csv', header=None)

num_nodes = adj_matrix.shape[0]
data = data.values.reshape(len(data), num_nodes, -1)
```

```
[3]: fig = plt.figure()
ax = plt.axes(projection='3d')
true_x = data[:,12,0]
true_y = data[:,12,1]
true_z = data[:,12,2]
ax.plot(true_x, true_y, true_z, label='Node 12')
ax.legend()
plt.show()
```



```
[12]: from math import exp
def dx_dt(t, x, y, z, neighbors, coupling_coeff):
    self_dynamics = lambda x, y, z: -0.707 * z - 0.909 * (x**3) + 2.519 * x
    inter_dynamics = lambda x_i, x_j: ((-0.182 * x_i) + 0.779) * (0.059 * (1 / (1 + exp(-x_j))) + 0.076)

    dx = self_dynamics(x, y, z)
    for neighbor in neighbors:
        dx += coupling_coeff * inter_dynamics(x, neighbor[t, 0])
    return dx
```

```
[17]: node_idx = 12
data_init = data[0, node_idx, :]
neighbors = data.reshape(data.shape[1], data.shape[0], data.shape[2])
coupling_coeff = adj_matrix[node_idx]

x_values = [data_init[0]]
y_values = data[:, node_idx, 1]
z_values = data[:, node_idx, 2]

for t in range(len(data)):
```

```

    x_next = x_values[t] + dx_dt(t, x_values[t], y_values[t], z_values[t],
↳neighbors, coupling_coeff)
    x_values.append(x_next)

x_values = np.array(x_values)

```

ValueError Traceback (most recent call last)

Cell In[17], line 14

```

    11     x_next = x_values[t] + dx_dt(t, x_values[t], y_values[t],
↳z_values[t], neighbors, coupling_coeff)
    12     x_values.append(x_next)
----> 14 x_values = np.array(x_values)

```

ValueError: setting an array element with a sequence. The requested array has a
↳inhomogeneous shape after 1 dimensions. The detected shape was (50000,) +
↳inhomogeneous part.

```

[16]: time = np.arange(0, len(data))
fig, ax = plt.subplots()
ax.plot(x_values.squeeze())
ax.plot(true_x)
ax.legend(['x_values', 'true_x'])
ax.set_xlabel('Time')
ax.set_ylabel('Values')
ax.set_title('Comparison of x_values and true_x')
plt.show()

```

AttributeError Traceback (most recent call last)

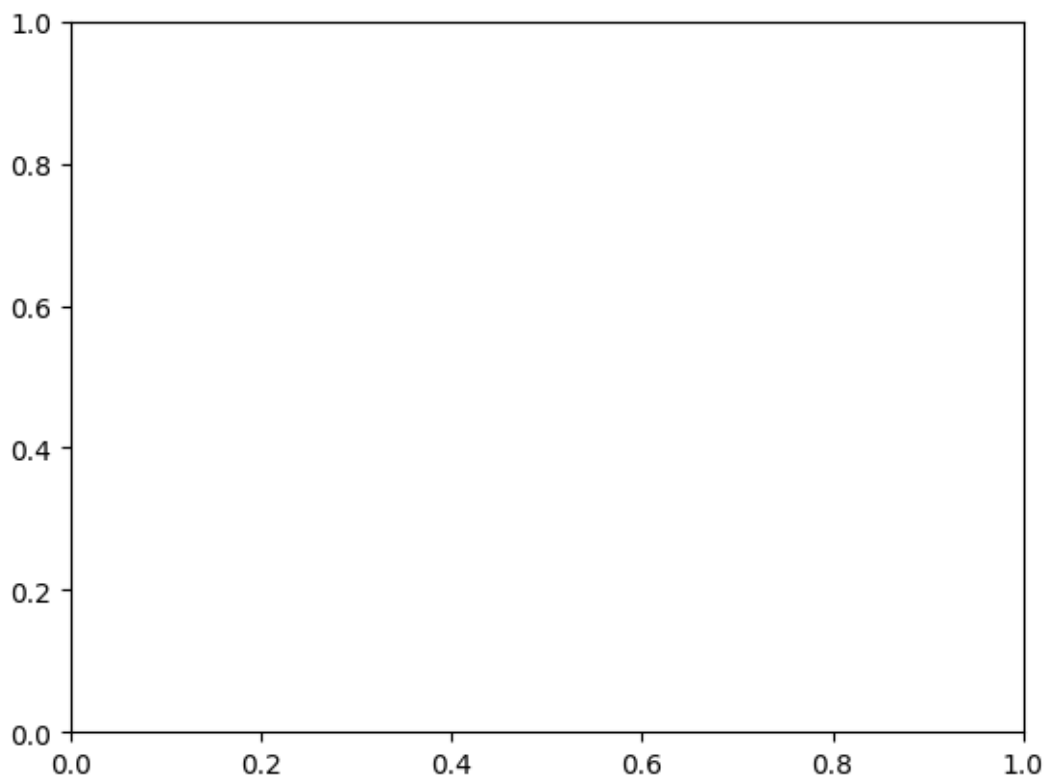
Cell In[16], line 3

```

    1 time = np.arange(0, len(data))
    2 fig, ax = plt.subplots()
----> 3 ax.plot(x_values.squeeze())
    4 ax.plot(true_x)
    5 ax.legend(['x_values', 'true_x'])

```

AttributeError: 'list' object has no attribute 'squeeze'



[]: